RELATIONAL DATABASES
*TABLES

Relational Database

- Standard relational databases enable users to manage predefined data relationships across multiple databases

- Popular examples of standard relational databases include
  - Microsoft SQL Server
  - Oracle Database
  - MySQL
  - IBM DB2.

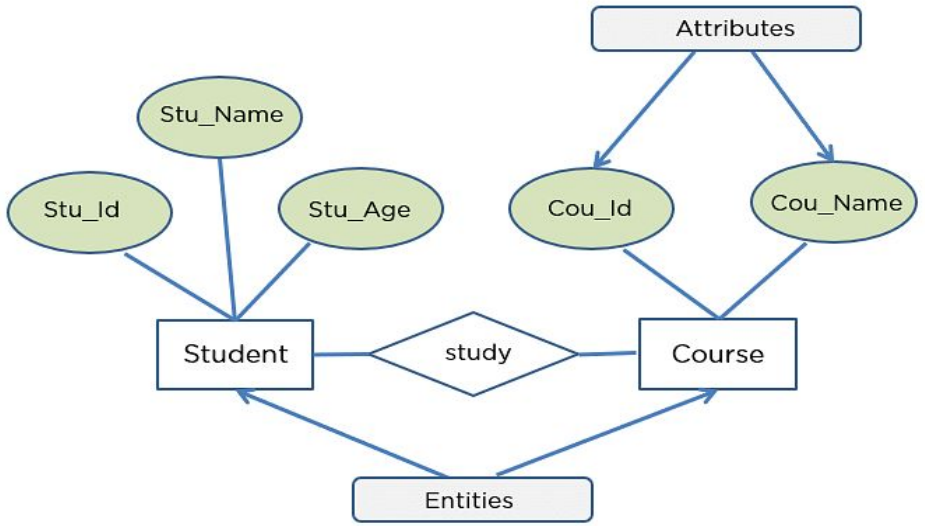
Terminologies in Relational Database

1. Entity - An entity in the database can be defined as **abstract data** that we save in our database. For example, a customer, products.

For example, Suppose we design a school database.

In this database, **the student will be an entity** with attributes like address, name, id, age, etc.

The **address can be another entity** with attributes like city, street name, pin code, etc and there will be a relationship between them.



Terminologies in Relational Database

2. Attributes - An attribute is a **detailed form of data** consisting of entities like length, name, price, etc.

3. Relationship - A relationship can be defined as **the connection between two entities** or figures. One Entity can be related to more than one entities.

For example, a person can relate to multiple persons in a family.

Terminologies in Relational Database

4. Primary key

- A primary key is the pointer of records that is **unique and not null** and is used to uniquely identify attributes of a table.
- A Primary Key is the **minimal set of attributes** of a table that has the task to **uniquely identify the rows**, or the tuples of the given particular table.

5. Tuple

- A **tuple** contains all the information of a particular entity. The table is composed of fields(columns) and a tuple(rows).
- A tuple represents **a row in the table and the data associated with the entity**.

The choice of the possible primary key from the candidate keys depend upon the following conditions.

Terminologies in Relational Database

Primary key

The choice of the possible primary key depends upon the following conditions.

- **Minimal:** The primary key should be composed of the minimum number of attributes that satisfy unique occurrences of the tuples.
- **Accessible:** The primary key used should be accessible by anyone who wants to use it. The user must easily insert, access or delete a tuple using it.
- **NON-NULL Value:** The primary key must have a non-null value for each tuple of the relation, which is required for the identification of the tuple.
- **Time InVariant:** The values of the primary key must not change or become null during the time of a relation.
- **Unique:** The value of the primary key must not be duplicated in any of the tuples of a relation

Terminologies in Relational Database

6. Foreign key - It acts as **a referral to the Primary Key of another table**. A foreign key contains columns with values that exist only in the primary key column they refer to.

A foreign key is the one that is used to **link two tables together via the primary key.**

It means the **columns of one table points to the primary key attribute of the other table**. If any attribute is set as a primary key attribute, it will work in another table as a foreign key attribute..

Terminologies in Relational Database

Example of Foreign Key

Consider two tables **Student** and **Department** having their respective attributes as shown in the below table structure:

Student

Stud_Id	Name	Course
101	John	Computer
105	Merry	AI
107	Sheero	Biology
108	Bisle	Maths

Department

Dept_name	Stud_Id
CS_Department	105
CS_Department	101
Science_Department	101
Math_Department	108

Terminologies in Relational Database

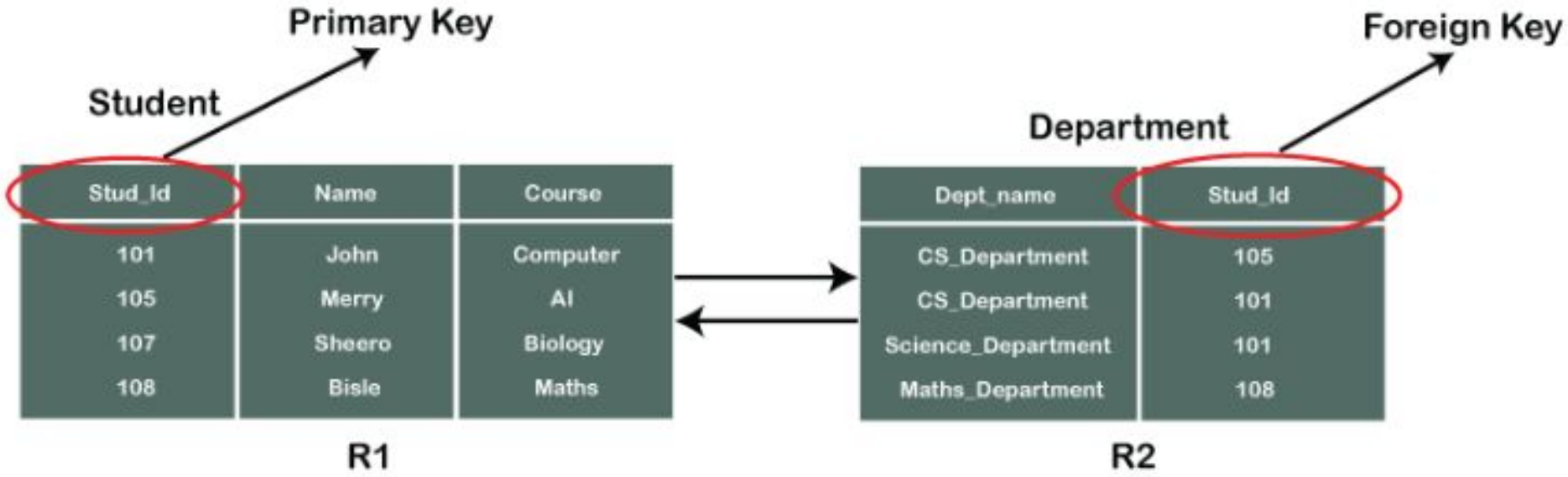
Example of Foreign Key

- In the two tables, one attribute is common, that is **Stud_Id**, but it has different key constraints for both tables.
- In the Student table, the field Stud_Id is a **primary key**.
- Stud_Id is a **foreign key** attribute for the Department table because it is acting as a primary key attribute for the Student table.
- It means that both the Student and Department table are **linked with one another** because of the Stud_Id attribute.

Terminologies in Relational Database

Example of Foreign Key

- The following diagram shows the structure of the relationship between the two tables



Terminologies in Relational Database

7. Normalization -

- A flexible data model needs to follow certain rules. Applying these rules is called normalizing

- The main reason for normalizing the relations is **removing insertion, deletion and updating anomalies**

- Failure to eliminate anomalies leads to **data redundancy** and can cause **data integrity** and other problems as the database grows

- Normalization consists of a series of guidelines that helps to guide the user in creating a good database structure.

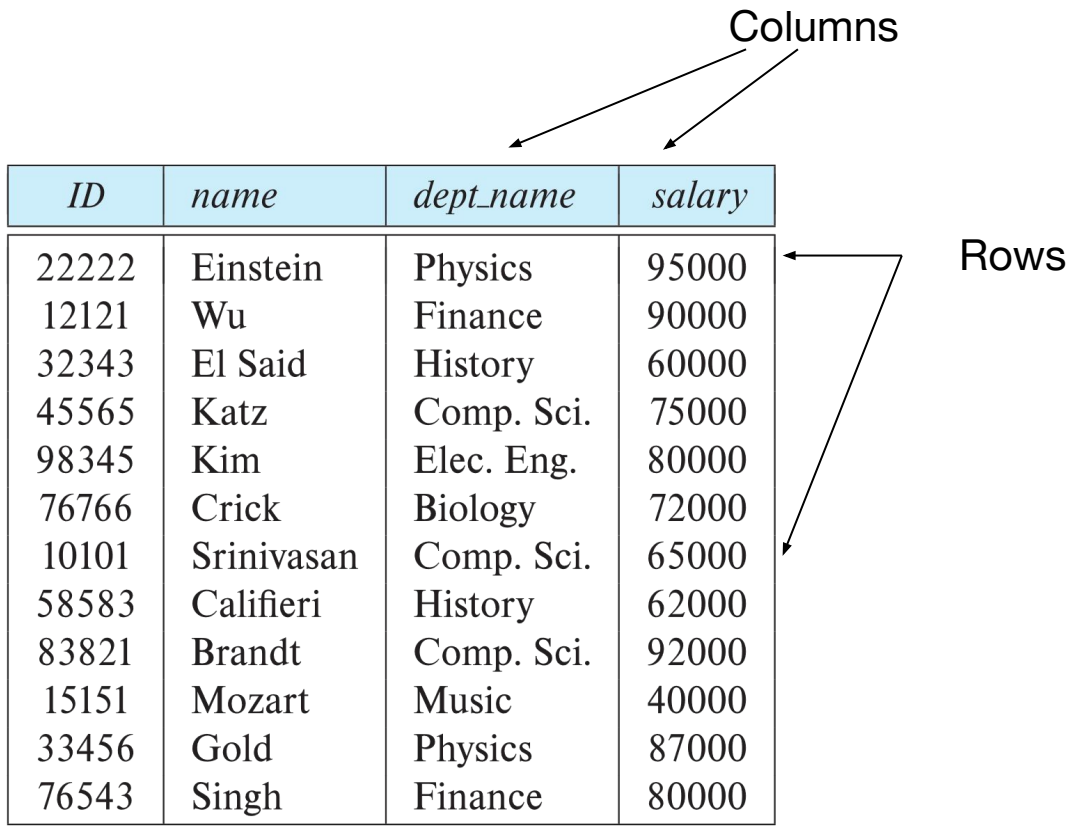
What is RDBMS (Relational Database Management System)

- **RDBMS** stands for *Relational Database Management System*
- All modern database management systems like SQL, MS SQL Server, IBM DB2, ORACLE, My-SQL, and Microsoft Access are based on RDBMS
- It is called Relational Database Management System (RDBMS) because it is based on the relational model introduced by E.F. Codd.

Tables

- Tables are database objects that contain all the data in a database
- In tables, data is logically organized in a **row-and-column format** similar to a spreadsheet
- Each row represents a **unique record**, and each column represents **a field (attribute)** in the record.
  - For example, a table that contains employee data for a company might contain
    - a row for each employee and
    - columns representing employee information such as employee number, name, address, job title, and home telephone number.

Tables



ID	*name*	*dept_name*	*salary*
22222	Einstein	Physics	95000
12121	Wu	Finance	90000
32343	El Said	History	60000
45565	Katz	Comp. Sci.	75000
98345	Kim	Elec. Eng.	80000
76766	Crick	Biology	72000
10101	Srinivasan	Comp. Sci.	65000
58583	Califieri	History	62000
83821	Brandt	Comp. Sci.	92000
15151	Mozart	Music	40000
33456	Gold	Physics	87000
76543	Singh	Finance	80000

(a) The *instructor* table